

Homework: Limits and Proofs

Due end of workshop.

Instructions: Do not use a calculator. Solve the following and type your work and answers.

1. Consider the function $f(x) = \frac{3x^2 + 4x - 4}{x^2 - 1}$, find the following and show their existence through limits.
 - (a) Find the horizontal asymptote. Write your answer as $y = b$.
 - (b) Find all the vertical asymptotes. Write your answer(s) as $x = a$.
2. (30 points) The Sandwich (or Squeeze) Theorem states that:
If
 - i. $g(x) \leq f(x) \leq h(x)$ for all x on some open interval containing c , except possibly at c ,
and
 - ii. $\lim_{x \rightarrow c} g(x) = L = \lim_{x \rightarrow c} h(x)$

Then,

$$\lim_{x \rightarrow c} f(x) = L$$

Use this theorem to prove that $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$.